

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	Sb-0000000005
Award Program	SMART Broiler
Project Title	OpticFlock: automated assessment of broiler chicken welfare that prioritizes animal welfare
Grant Start Date	01/01/2020
Grant End Date	12/31/2021 (with 6 month extension)
Total Grant Budget	\$232062.00
Total FFAR Budget	\$232062.00
Matching Funder(s)	
Final Report Date	09/12/2022
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General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.
Springdale, Arkansas, 3rd Arkansas Congressional District, USA.
- 1.2. How many new jobs were created because of this grant funding?
0.
- 1.3. How many jobs were maintained because of this grant funding?
2.
- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.
No.
- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.
The University of Oxford has not changed but the Dept. Zoology has become the Dept. of Biology, so the address is now University of Oxford, Department of Biology.

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Across the world, over 70 billion chickens are reared for meat each year and there is growing public concern over the welfare of these birds. In response to this concern, we have developed a camera surveillance system, OpticFlock, that continuously monitors chicken behavior throughout their lives and delivers a welfare 'verdict' every 15 minutes. OpticFlock uses the movement patterns made by chicken flocks as they move around the house

and can detect welfare problems, such as lameness or disease, by changes in these movement patterns often before they are obvious to the human eye. By providing constant welfare monitoring and early warning of health and welfare issues, the aim of OpticFlock is to improve the overall welfare of broiler chickens by enabling farmers to manage their flocks to welfare higher standards.

During Phase 1 of the project, we successfully completed our main objective, which was to validate the use of OpticFlock on US broilers (we had previously only used it on European farms). Despite delays due to COVID, our cameras collected data from a total of 20 flocks at the Tyson Research Farm in Arkansas. On those same flocks we also have data on humidity, internal and external temperature, water use, body weight, mortality, infections, ‘paws’ and (through collaboration with Audio-T) data on sounds in the houses. Our results from Tyson combined with work from elsewhere show that OpticFlock is able to detect the following key welfare indicators: mortality, weak birds, respiratory infection, hockburn (skin damage to legs), lameness and walking ability. A key finding was that OpticFlock detects problems when the birds are very young (less than a week old), emphasizing the importance of this period of their lives for future welfare and providing an important early warning of problems to producers.

We were also able to make considerable progress with our second Phase 1 objective – to develop a commercially-ready version of OpticFlock so that it would be taken up and used by producers. Farm managers are busy people and if equipment, however potentially useful it may be, is difficult to set up or keeps breaking down, they will not use it. For full commercialization, OpticFlock needs to be a robust, user-friendly, plug-and-play system. We have been able to improve OpticFlock so that it is now much easier to install, can restart itself without losing data in the event of a power outage and constantly monitors itself to make sure all its elements are working properly.

Our third aim was to integrate the information from OpticFlock with other sorts of information such as temperature, humidity, water use, sound etc. We have shown that there is a high correlation between sound and the movement patterns of flocks and are continuing to work on combining all different sorts of information so that producers will have a fully integrated flock management system that includes all aspects of the health, environment, welfare and behavior of their flocks.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

Our aim is to have a welfare monitoring system that can be widely used on commercial broiler farms. At a time when there is increasing public concern about the welfare of chickens kept for meat and major threats to the poultry industry from rising food prices, heating costs and changing food habits by the public, the industry is under pressures to improve both welfare and efficiency of production, made even more challenging by pressures to achieve this without the use of antibiotics and at the same time to reduce emissions, water and land use. OpticFlock aims to help the poultry industry to meet these challenges by providing an inexpensive, easy to use way of potentially increasing both welfare and efficiency. First, it is designed as a welfare monitor – that is, its CCTV cameras keep flocks under surveillance continuously throughout the whole life of a flock. Small on-farm computers immediately process the camera images and turn them into a numerical welfare index that appears on a screen. No images (only numbers) leave the farm so the system is very secure and does not intrude into the privacy of farm staff. We have already demonstrated (from trials in the UK, Switzerland and the US) that OpticFlock can detect key welfare outcomes such as mortality, hockburn, lameness as well as signs of infection such as respiratory disease, *Campylobacter* etc. Our key finding has been that OpticFlock detects these problems *very early* in the life of a flock, often when the birds are less than a week old, before clinical signs appear and before the human eye can see that there is anything wrong. This means that it has the

potential to make a major contribution to flock management by providing the farm manager with early warning of problems and giving time for interventions to prevent those problems from becoming more serious. This is particularly important where prophylactic medication is not being used and the health of a flock is even more dependent on good management. Monitoring welfare allows producers to reassure their customers that they give high priority to welfare and are prepared to show what standards they are putting in practice. This potentially brings the whole industry more in line with public values.

Second, OpticFlock potentially brings financial advantages through greater efficiency and less waste. High levels of mortality, having to cull birds because of walking difficulties and downgrades all cost money. If OpticFlock can help to reduce mortality by even 1%, the savings across the industry would be significant, there would be less food wasted and less environmental impact. Healthier birds means less need for medication, a healthier environment for farm staff to work in and safer end food products.

Although OpticFlock has performed consistently well in research trials across two continents, there are still a number of hurdles to overcome before it will be fit to go on sale as a commercial product. A chicken house is dusty and humid, is sometimes subjected to fumigation and power-washing and may frequently be liable to fluctuations and outages in electrical power and in internet connections – not an ideal environment for electronic equipment. To be fit for sale, it would need to satisfy electrical and other standards that may differ from country to country. For farm managers to want to use it, it needs to be simple to use, easy to install and not liable to break down. Its data has to be presented in a form that is most useful for farmers – easy to interpret and easy to integrate with other information. Each of these problems may sound trivial when considered in isolation but together they explain why Optic Flock is not yet market-ready. We are gradually working through each of the problems we encounter and seeing each solution we find as a step towards a marketable product. For example, we are in process of transferring the software onto an industrial grade, rugged Raspberry Pi which we hope will make the system better able to function on commercial farms and solve some of the testing and standards issues. We have introduced software that enables Optic Flock to monitor itself and check that everything is working. We have introduced more software to deal with power outages so that no data are lost and the system can restart itself. We are aiming to have two versions of OpticFlock - a stand-alone version that can be used without any other equipment and a separate version designed to integrate with existing dashboards used by control systems companies.

Despite the fact that the findings from this project cannot yet be applied in practice for the reasons given above, OpticFlock has already proved itself as a welfare monitoring tool with great potential for helping the poultry industry with the many challenges it currently faces. Its ability to detect, and often predict, welfare outcomes shows that it has already passed the proof-of-concept stage and has a future as a commercial product. If developed further, it has the potential to ensure the higher standards of farm animal welfare that are being demanded by the public. It could help farmers deliver safe, affordable food and still meet new environmental requirements. By providing easier flock management it could deliver early warning of problems and so help producers succeed without the aid of antibiotics or other medication. It has a role in the future planning of the poultry industry through its potential to bring benefit to society, to farmers and to animals themselves.

2.3 Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that

- could impact U.S. agriculture and food systems, if used;
- b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Potential outcomes

OpticFlock is still at the stage where it has shown its potential for helping the poultry industry but it is still being developed as a commercial product. Trials on over 300 flocks in Europe and the US have shown that it has great potential as a way of monitoring the health and welfare of broiler chickens throughout their lives and could impact world agriculture in the following ways:

- Inexpensive and continuous welfare monitoring leading to higher welfare standards
- Early warning of health and welfare problems leading to targeting of medication and lower use of antibiotics
- Earlier treatment leading to lower mortality and so resulting in greater efficiency and less waste
- Evidence-base for welfare standards leading to greater public confidence in the poultry industry
- Round-the-clock flock monitoring giving easier flock management for farm staff
- Objective welfare judgements on effects of proposed innovations e.g. enrichments, light changes

2.4 What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

The goals of the project were:

- (i) Using the facilities of the Tyson Research Farm in Arkansas, our primary aim was to validate OpticFlock and demonstrate its applicability to US conditions (different breed and management from Europe). We also aimed to fill what was (and still is) a major gap in smart farming welfare assessment – namely, providing evidence that it can actually improve chicken welfare in practice. We planned to do this by showing that houses where OpticFlock information was available to the farm manager during the rearing of a flock could achieve higher standards of welfare than houses where no such information was available.
- (ii) Our second aim was to take steps towards developing a commercially-ready version (or versions) of OpticFlock by using feedback from the trial to improve its operation in a commercial farm environment, for example by improving the software so that it could save data and restart itself in the event of a power outage, would be easy to set up and install etc. Each improvement was to be tested before the next step was taken to be sure it worked in farm conditions.
- (iii) Our third aim was to plan ahead for the future, so that after Phase 1, we would have a clear pathway to commercialization that would include ways of integrating OpticFlock data with other sorts of information, for example, temperature, humidity, sound etc.
- (iv) Our 4th aim was to collect enough data on OpticFlock to convince the Munters' Corporation that it would be in their commercial interest to develop it commercially.

The milestones for the project were:

Quarter 1 January 2020-March 2020.

Milestone 1. All details of trial finalised, including welfare audits. MD and SE to visit Tyson research farm

in Arkansas. ACCOMPLISHED: Regular online meetings with the Tyson team enabled the research plans to be finalised but visit by MD and SE prevented by Covid-19. Exchange of data agreed.

Milestone 2. OpticFlock equipment set up and running on trial farm. Delayed as components not delivered from China (due to COVID-19)

Milestone 3. Trial started in 4 houses at Tyson research farm. Delayed due to non-arrival of components

Milestone 4. Equipment installed and running at Munters' farm. Also delayed due to non-arrival of components.

ACCOMPLISHED: Discussions with Munters' over Development licence. One set of equipment shipped to Munters' Israel.

Milestone 5. Progress/problems discussed through Skype meetings. ACCOMPLISHED: Regular Zoom meetings with Tyson to finalise terms of Research Agreement and trial design. Zoom meetings with Jim Hollister and Lisa Sato at Munters'.

Quarter 2 April 2020- June 2020.

Milestone 1. First exchange of data from 4 houses at research farm (4 flocks). Still delayed due to the non-arrival of components.

Milestone 2. Database set up in Oxford in preparation for full data set.

ACCOMPLISHED: Research File Service facility discussed with the University of Oxford for data transfer and storage.

Milestone 3. All arrangements for smooth transfer, cleaning and storage of different sorts of data in place, together with security and access arrangements. ACCOMPLISHED: Research Agreement including data management signed with Tyson.

Milestone 4. Skype meetings with Tyson and with Munters'/OCC.

ACCOMPLISHED Regular (every one or two weeks) Zoom meetings took place with the Tyson technical team.

Instruction set-up manual written.

Milestone 4. Online discussions with both Tyson and Munters' continued.

ACCOMPLISHED: Further discussions took place between Munters' and Oxford University Innovation about the details of the Development Licence.

Quarter 3 July 2020-September 2020.

Milestone 1. Next exchange of data (expect to have 8 flocks completed). Delayed for reasons given above.

ACCOMPLISHED: With components now delivered, 8 OpticFlock units assembled in the UK, tested and shipped to Tyson with full instruction manual.

Milestone 2. Preliminary scrutiny of results in preparation for analyzing full results.

ACCOMPLISHED: Tyson team installed first OpticFlock units on farm.

Milestone 3. Business plan for full commercialisation of OpticFlock completed, including licensing arrangements.

ACCOMPLISHED: Provisional Development License agreed with Munters and sent to University of Oxford legal team.

Milestone 4. Skype meeting with Tyson to discuss progress.

ACCOMPLISHED: Regular meetings including on-line trouble-shooting with camera set-up.

Quarter 4 October 2020- December 2020.

Milestone 1. Next exchange of data on flocks complete to date (12 flocks). Delayed.

ACCOMPLISHED: Planned start of trial on 25th October unsuccessful because of technical problems. Technical assistance given through OCC with successful resolution.

Milestone 2. First draft of instruction manual written. ACCOMPLISHED: Full description and instructions already being used by Tyson.

Milestone 3. First updated algorithm written. Delayed and now planned for January/February 2021.

Milestone 4. Skype reporting meetings with Tyson. ACCOMPLISHED: Zoom meetings every two weeks in preparation for trial start date on 14th December.

Quarter 5 January 2021-March 2021.

Milestone 1. Complete analysis of research trial data received so far (estimated data from 16 flocks. Owing to delays listed above, this was not achieved.

ACCOMPLISHED: successful data collection from first four Tyson flocks

Milestone 2. companies (including Perdue, Avara) contacted to plan Phase 2. Application for Phase 2 to FFAR planned. Delayed.

Milestone 3. 24 Research trials completed. ACCOMPLISHED: 8 research trials completed.

Milestone 4. All data exchanged, cleaned and stored. More advanced analysis of data and development of new algorithms. Not possible. ACCOMPLISHED: Software up grade to allow system to deal with power outages without losing data.

No-cost extension applied for to compensate for delays.

Quarter 6 April 2021-June 2021.

Milestone 1. All data analysis completed. ACCOMPLISHED: 12 research trials completed

Milestone 2. Final report written and scientific output prepared for publication.

Milestone 3. Application for Phase 2 funding submitted. Delayed. ACCOMPLISHED: No cost extension granted.

Quarter 7 July 2021-September 2021 (Into no-cost extension period).

ACCOMPLISHED: 16 research trials completed. Analysis of data so far completed.

Quarter 8 October 2021- December 2021.

ACCOMPLISHED: 24 research trials completed and analyzed. Pre-application for Phase 2 funding submitted.

Successful negotiations with Tyson to continue collecting data after the end of the Phase 1 FFAR grant.

2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit).

The following change has occurred:

The original aim (see section 2.4 above) was to demonstrate that using OpticFlock actually increased the welfare of a flock, by directly comparing flocks when a farm manager had access to Opticflock data during the life of a flock with similar flocks when this live information was not available. However, this proved not to be possible because many other experiments were being conducted in the same 4 houses at the Tyson Research Farm (including other FFAR projects) so any comparison between houses would have been confounded by many other differences such as breed, lighting etc.

2.6 What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit).

1) *Major activities*. The major activity for this project was the setting up of OpticFlock (2 units/house) at the Tyson Research Farm in Arkansas and the successful collection of data from 24 different broiler flocks. These data were then put together with data supplied by Tyson on management (e.g. water use, feed consumption, temperature etc.) and outcome measures (mortality, culls, body weights etc.) supplied by Tyson on each of these same flocks. This combined data set allowed us to achieve our main objective of validating Optic Flock for use on US farms. This was the first time that OpticFlock had been remotely installed – that is, set up by someone unfamiliar with it and following our instructions (and help over the internet). This in itself was an important activity as it taught us a lot about the instructions needed by someone new to the equipment, what they found easy, what needed to be explained more fully and the best way to prepare and pre-load the equipment for easy deployment. Ease of installation is essential for OpticFlock becoming a commercial product so our thwarted aim of not being able to visit the Tyson farm to set up our equipment because of Covid and having to rely on the farm staff there had the unexpected benefit that we had to improve our installation manual and make everything as easy as possible to set-up. We now have a detailed and comprehensive manual as well as improved set-up software and remote access.

Another major activity has been data analysis, which is still ongoing as we have only just been given access to a very important data set. Our preliminary analysis (see below) was based on our own camera data and welfare outcomes such as mortality, number of weak birds etc. supplied by Tyson. Since the end of Phase 1, been in touch with the Munters' Corporation who have their Sonar equipment on this farm and they have given us access to very detailed data on the same flocks (internal and external temperature, humidity, water use etc. every 15 minutes, giving us a great opportunity to combine OpticFlock data with environmental and other data on a much finer time scale. We are currently analyzing this data. In addition, we have collaborated with Audio-T and exchanged data with them on the same flocks. There is a high correlation between their sound data and our flock behavior data.

2) *Specific Objectives* (Please see section 2.4 above). We were able to achieve our first goal of validating OpticFlock for use on US farms by the specific objective of collecting Opticflock data on 24 flocks and showing that OpticFlock gave comparable results to those obtained on European farms, despite obvious differences such as light regimes etc. Towards our second goal of preparing OpticFlock for the commercial market we achieved the following objectives:

- Development of software for self-monitoring so that OpticFlock regularly checks that all cameras are working, all required data is being received etc.
- Development of software to deal with power outages so that data is not lost and the system restarts.
- Adjustment of the graphical output to incorporate local conditions
- Writing a comprehensive manual for OpticFlock including description, installation and trouble-shooting
- Setting up remote monitoring so that we can see data in real time from Oxford
- Provision for online data transfer from Tyson to Oxford and permanent storage
- Upgrading of the software to make it more stable and easier to install
- Development license signed with Munters' for the commercial development of OpticFlock.

3) *Results*

Our two most significant results are (i) that OpticFlock can detect flocks with reduced levels of welfare (as measured by mortality, no. weak birds etc.) and (ii) that it can do so in very young birds, making it making it potentially a valuable a flock management tool able to give early warning of problems.

OpticFlock currently detects flocks with the following indicators of poor welfare:

- % final mortality: correlated with OpticFlock data from 4 days, with high levels of significance ($p < 0.001$), validated on over 300 flocks (45 in Switzerland, 20 in Denmark, 230+ in UK), c.100 birds/flock captured on video. Tyson flocks showed correlation with OpticFlock at 2 days with $r = 0.81$, $p < 0.005$.
- % 'weak' birds (unknown infection, often *E.coli*) Tyson flocks showed correlation with OpticFlock data, $r = 0.71$, $p < 0.01$ by 6 days of age.

- % final hockburn (predictable from 4 days): 280+ flocks (UK, Denmark, Switzerland) Significant correlations found in all studies from 4 days of age. Data on hockburn was not available for Tyson flocks, but we hope to measure in Phase II.
- *Campylobacter* infection: predictable from 7 days: 31 flocks (UK)
- Lameness and difficulty walking: predictable from 21 days. 44 flocks (UK and Switzerland). At 2 weeks, OpticFlock welfare measure at flock level was significantly correlated with walking ability of individuals, 20 flocks, 16 individuals/flock $r=0.081$, $p<0.01$.

As of now, OpticFlock does not distinguish between these welfare measures i.e. it cannot yet say that flock X has a high % of lame birds but is free from infection whereas flock Y is infected with *Salmonella* but has no leg problems. This may be because these different welfare indicators often go together or it may be that as we collect more information, it will be possible to identify the movement characteristics of flocks with different classes.

With the large numbers of chickens that are kept together (often 30,000+), tracking individual birds is currently computationally difficult and we have shown that measuring overall flock movement provides useful information that is computationally simple enough to operate in real time and so give a continuously updated welfare verdict. Furthermore, a farm manager does not want an individual welfare report on each of 30,000 birds as he or she has to take decisions that affect the whole flock. Factors such as lighting regimes, vaccinations, drinker height, diet are not adjusted to the needs of individual birds. They are taken at flock level and so the farmer needs flock level information. OpticFlock provides this.

In the future, we may see the development of ‘precision’ broiler farming, comparable to precision crop agriculture where plots or even plants are given their own individual treatment, but as of now, broiler chickens are farmed as a flock and therefore assessing their welfare as a flock is the most important first step to take. However, although OpticFlock works at flock level we have shown that it’s output is *correlated* with individual bird welfare (see above).

Another future line of research that will be key to the commercial take-up of OpticFlock will be to demonstrate that its use actually does improve standards of chicken welfare. We had planned to do this as part of this project by comparing the welfare standards of houses with and without live OpticFlock data available to the farm manager but this was not possible due to other studies being carried out in the same houses, giving too many confounding variable for valid conclusions to be drawn.

4) *Key outcomes*. A Development License with the Munters’ Corporation was signed. Although this does not now seem to us the best pathway to commercial development, it does show that there is industrial interest in Opticflock.

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

Running trials on commercial farms is the best way to develop a system that is relevant to the farming community but this does mean that there is less control over what happens than with smaller scale pen trials. For example, on the Tyson farm, although in many ways ideal for carrying out research, several different experiments were being conducted at the same time as our trials so that different houses might have different breeds, different lighting regimes or enrichments etc. For this reason we decided it would not be valid to carry out our stated objective to compare houses with and without OpticFlock as there would be too many confounding variables. Instead, we recorded as much information about each house as was available and were then careful about the inferences we drew from this data. Some factors, such as season of the year can be dealt with by matching flocks at different times of year, but others such as outbreaks of disease are completely beyond our control. Professor Christl Donnelly has provided expert statistical advice at all stages.

2.8. List key stakeholders that could be served by results of this project.

- Farm staff, including managers
- Chicken producer companies
- Food outlets, restaurants
- Food retailers
- Consumers of chicken and chicken products
- Manufacturers of heating and ventilation equipment for chicken houses
- Concerned members of the public
- NGOs, legislators and politicians
- Researchers

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write “nothing to report.” A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Nothing to report (yet)

2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

1. Although we ordered components from China before the grant officially began, there was a delay in receiving these as production in China had shut down during the COVID-19 pandemic. This meant that we could not assemble the equipment and so could not send it to the Tyson research farm. This caused a considerable (up to 4 months delay).
2. When we did finally receive all the necessary components and were able to assemble the 8 units needed for the Tyson trial, we found that the cameras been ordered (which the manufacturers had assured us were the same model as before) did not work. We had then to track down the source of the problem (the manufacturer had in fact made some minor changes) and a software modification had to be made to accommodate this. This caused a delay while the software was re-written and tested but will be useful for future installations.
3. The COVID lockdown in the UK meant that there were delays in getting the equipment collected for shipping. It was then delayed by US customs so there was a gap of over a month between the equipment being ready to go and finally arriving in Arkansas.

4. The Tyson team have been fantastic in setting up the equipment but they initially could not get it to work, despite our very extensive instructions. We solved this by on-line trouble-shooting sessions with our software developers, OCC. The lessons we learnt from this initial set-up (the first we have done remotely) were then used to modify the instruction manual to make it more user-friendly. This will be important for future commercial use of OpticFlock.

We also used the experience of the Tyson team to draw up a list of improvements, both software and hardware, that would make OpticFlock easier to use and more useful for farmers.

2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit).

It has provided data for student projects (via Audio-T) – see below

2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

3 undergraduate students worked on our data and a post-doc. Dr. Sarah Hayes is about to start work.

3. Information Products

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award. *None yet.*

3.2. Please provide a list of citations for the information products produced during the project. *None yet*

3.3. *Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during*

theproject resulting directly from the FFAR award? If yes, please provide citation. No.

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

OpticFlock has appeared as a case study on the website of our software developer, Oxford Computer Consultants: <https://www.oxfordcc.co.uk/customsoftware/opticflock-helping-chicken-producers-to-achieve-welfare-and-efficiency/>

It is also described, together with the research underpinning it, on <http://users.ox.ac.uk/~snikwad/styled3/index.html>

- 3.5. *Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s). No.*

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit).

The Development License with Munters' runs until November 2022. The documentation describing OpticFlock in detail and how it works is part of the IP arrangements with Munters and will remain confidential under the Development Licence agreement.

The instruction manual will be publicly available once we are satisfied that it is complete and as helpful as possible.

- 3.7. Beyond depositing information products in a repository, what other activities *have you*

undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

Farm data is confidential and we understand that it is the property of the farmer or company. We therefore do not own the data and have an NDA in place with Tyson Foods. Any requests for information would therefore have to be agreed with Tyson.

4. Data Management

4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

- Optical flow data for 24 flocks – spreadsheets
- Production data for 24 flocks – excel spreadsheets
- Environmental data for 24 flocks

4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

The data sets all refer to the same 24 flocks

4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

This will only be possible with the agreement of the owners of the data.

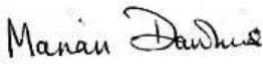


Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Marian Stamp Dawkins

PI Signature:  Date: 28th Sept. 2022

Authorized Signing Official (ASO) Name: Ana Serra Berros

ASO Signature: Please see official signature page Date: 12th October 2022